

From-Scratch CNNs vs. Transfer Learning for Plant Disease Classification: An Investigation of Intrinsic Disease Confusions

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Abstract—Classifying plant diseases from photographs of leaves is a task that has been studied many times, and modern neural networks now reach very high accuracy on it. However, most of this work reports a single accuracy number and stops there, which tells us little about which diseases a model still gets wrong, or why it gets them wrong. In this paper we look more closely at that question. Working with a balanced ten-class subset of the PlantVillage dataset that covers four crops (tomato, corn, grape and potato), we train a convolutional neural network from scratch, we fine-tune a ResNet-18 that was pretrained on ImageNet, and we retrain the from-scratch network with data augmentation so that we can compare the three settings fairly. We evaluate every model on a held-out test set and study its confusion matrix and its per-class scores rather than its accuracy alone. Transfer learning raised test accuracy from 91.0% to 97.8%, an improvement of 6.8 percentage points, while augmentation added only about one point and mainly reduced overfitting. The most interesting result is that the confusion between tomato early blight and tomato late blight was the hardest error to remove: these two classes had the lowest F1 scores in every model, even after transfer learning had cleared up almost every other confusion. This suggests that the two diseases simply look alike in these images, rather than the confusion being a weakness that a stronger model could fix. We discuss what this means for how such models should be evaluated and deployed.

Index Terms—plant disease classification, convolutional neural networks, transfer learning, data augmentation, model evaluation, PlantVillage.

I. INTRODUCTION

I have always been fond of growing plants, and tending a small garden has been a quiet source of enjoyment for me. Gardening and time spent around greenery are commonly associated with better mood and general wellbeing, which is part of why so many people keep plants at home. Tomatoes are among the easiest plants to grow, and for that reason they are a common first choice, but they are also unusually prone to pests and infections. I learned this directly. I was growing about ten tomato plants, and one of them developed a blight that I did not recognise. Because I could not identify it in time, it spread and eventually ruined most of my small crop. That experience is what drew me to the problem studied in this paper: whether a machine could tell such diseases apart from

a photograph, and, just as importantly, whether some diseases are simply too similar in appearance to be separated reliably.

Crop diseases are a serious and lasting threat to the world's food supply, and being able to identify a disease early, from nothing more than a picture of a leaf, can help a grower act before the problem spreads [1], [3]. Diagnosing a disease by eye usually requires expert knowledge, which is not always available to small farmers, and this is the main reason so much effort has gone into building automatic systems that recognise plant diseases from images [1].

Convolutional neural networks are now the usual tool for this task, and on the widely used PlantVillage dataset many systems report accuracies above ninety-nine per cent [1], [7]. A single high accuracy figure can be misleading, though, because it hides a more useful question: which diseases does the model still confuse, and is that confusion the fault of the model or a property of the diseases themselves? Most earlier work has aimed at pushing the accuracy number up and has not tried to separate these two explanations.

The distinction matters in practice. If a model keeps confusing two diseases because the model is too weak, then a better network or more data should fix it. But if the two diseases genuinely look alike in the image, then no amount of extra model capacity will help, and a real diagnostic tool would be better off admitting its uncertainty than giving a confident but unreliable answer. Telling these two cases apart cannot be done by looking at one accurate model in isolation; it needs a controlled comparison. That is what we set out to provide.

In this study we treat the question of residual confusion as the main object of interest. We use a deliberately balanced ten-class subset of PlantVillage that includes several pairs of visually similar diseases, and we compare three training settings under identical data. The first is a convolutional network trained from scratch. The second is a ResNet-18 [2] that was pretrained on ImageNet and then fine-tuned on our data. The third is the same from-scratch network, retrained with data augmentation. Instead of reporting only accuracy, we examine the per-class scores and the confusion matrices so that we can follow how individual confusions behave as the training setting changes. Our main contributions are a

controlled measurement of how much transfer learning helps compared with augmentation, an observation that augmentation acts here mainly as a way to reduce overfitting rather than to raise accuracy, and the finding that the tomato early-versus-late blight confusion is the most stubborn error of all and persists across every model we tried.

II. RELATED WORK

The use of deep learning for plant disease recognition on PlantVillage was made popular by Mohanty *et al.* [1], who trained deep networks across all thirty-eight classes and reported accuracies above ninety-nine per cent under controlled conditions. A good deal of later work has continued to push these numbers higher using newer and more elaborate architectures, including transfer-learning approaches based on EfficientNet [7]. Recent surveys trace the whole progression, from classical machine learning through convolutional networks to vision transformers.

A recurring idea in this literature is transfer learning: rather than training from scratch, one starts from a network already trained on the very large ImageNet dataset [2], [6], because the general visual features it has already learned give a strong starting point when task-specific data are limited. Regularisation techniques such as dropout [5] and data augmentation are likewise standard for improving how well a model generalises. The work of Wang *et al.* [3] is particularly relevant to us, because it compared a shallow network trained from scratch against a fine-tuned pretrained network for estimating disease severity, reaching about ninety per cent on an apple black-rot task. We adopt a similar comparative approach, but we apply it to multi-crop disease identification and, unlike most of this earlier work, we make the analysis of residual confusions the central point rather than an afterthought.

III. DATASET

We use the publicly available PlantVillage dataset [1], which holds more than fifty-four thousand colour images of leaves, spanning fourteen crops and thirty-eight classes, all photographed under controlled conditions. Rather than use the whole dataset, we chose a balanced subset of ten classes across four crops, and we did so with three aims in mind. First, we wanted the classes to be balanced: every class we selected has at least a thousand images available, and we cap each one at six hundred so that no class can dominate simply by being larger. Second, we wanted several crops rather than one, so that the model has to learn both which crop it is looking at and which disease is present. Third, and most deliberately, we included pairs of diseases that are known to look similar, namely early and late blight in both tomato and potato, so that we would have a meaningful basis for studying where the models fail. The ten classes are listed in Table I.

The six thousand images were divided into a training set of four thousand two hundred, a validation set of nine hundred, and a test set of nine hundred, using a seventy-fifteen-fifteen split. The split was made once with a fixed random seed so

TABLE I
SELECTED TEN-CLASS BALANCED SUBSET (600 IMAGES PER CLASS USED).

Crop	Class	Avail. images
Tomato	Healthy	1591
Tomato	Early blight	1000
Tomato	Late blight	1909
Tomato	Bacterial spot	2127
Corn	Common rust	1192
Corn	Healthy	1162
Grape	Black rot	1180
Grape	Esca (Black Measles)	1383
Potato	Early blight	1000
Potato	Late blight	1000
Total used (600 × 10)		6000

that it can be reproduced exactly, and the test set was set aside and used only once, for the final results reported here.

IV. METHODOLOGY

A. Preprocessing

Every image was resized to 128×128 pixels, converted into a tensor with pixel values scaled to the range zero to one, and then normalised using the standard ImageNet channel statistics so that the pretrained network would behave as expected. In the augmentation setting, the training images were additionally altered at random during training by horizontal flips, small rotations of up to twenty degrees, and mild changes in brightness and contrast. The validation and test images were never augmented.

B. From-Scratch Network

Our baseline network is a small convolutional model with three convolutional blocks, in which the number of channels grows from three to sixteen, then to thirty-two, and then to sixty-four. Each block uses a three-by-three convolution with padding, a ReLU activation, and two-by-two max-pooling. After the convolutional part, the features are flattened and passed through two fully connected layers, reducing from sixteen thousand three hundred and eighty-four features to one hundred and twenty-eight and then to ten outputs, with dropout applied before the final layer [5]. Altogether the model has about two point one million trainable parameters.

C. Transfer Learning

For the transfer-learning setting we used ResNet-18 [2], which had been pretrained on ImageNet [6]. We replaced its original thousand-way output layer with a fresh ten-way layer and then fine-tuned all of the network’s weights on our leaf images.

D. Training

All of the models were trained using the Adam optimiser [4] together with the cross-entropy loss. The from-scratch models were trained for twelve epochs with a learning rate of 10^{-3} , while the fine-tuned ResNet was trained for eight epochs with a smaller learning rate of 10^{-4} , reflecting the fact that it was

TABLE II
TEST-SET ACCURACY ACROSS TRAINING REGIMES.

Model	Test accuracy
From-scratch CNN (no augmentation)	91.0%
From-scratch CNN (+ augmentation)	92.0%
Fine-tuned ResNet-18 (transfer)	97.8%

only adjusting a network that had already learned a great deal. A batch size of thirty-two was used throughout, and all training was carried out on a single NVIDIA T4 GPU.

E. Evaluation

We report the accuracy on the held-out test set, the precision, recall and F1 score for each class, and the full confusion matrix for each model. To study the tomato blight confusion in particular, we also counted, separately for each model, how many tomato early-blight images were labelled as late blight and how many late-blight images were labelled as early blight.

V. RESULTS

A. Overall Accuracy

The test accuracies of the three models are summarised in Table II. Transfer learning gave by far the largest improvement, raising accuracy by almost seven points over the from-scratch baseline, whereas augmentation added only about one point.

B. Learning Behaviour and Overfitting

Fig. 1 shows how the loss and accuracy of the from-scratch model changed over training. For the first several epochs the training and validation loss fell together, but after that point the validation loss started to drift upward while the training loss kept falling, which is the usual sign of mild overfitting; dropout kept this from becoming severe. When augmentation was added, this gap disappeared: in the augmented model the training accuracy (0.901) actually ended slightly below the validation accuracy (0.918), which tells us that the main effect of augmentation here was to reduce overfitting rather than to raise the accuracy itself.

C. Confusion Matrices

The confusion matrices for the three models are shown in Figs. 2–4. The classes that look clearly different from the others, such as corn common rust and healthy corn, were classified almost perfectly by every model. The from-scratch model struggled most with the visually similar tomato disease classes, whereas transfer learning removed most of these errors.

D. Per-Class Performance

The F1 score for each class is given in Table III. In all three models the two lowest F1 scores belonged to the tomato blight classes, while the corn and potato classes were classified almost perfectly. Transfer learning raised the tomato blight scores considerably, but even so these remained among the weakest classes.

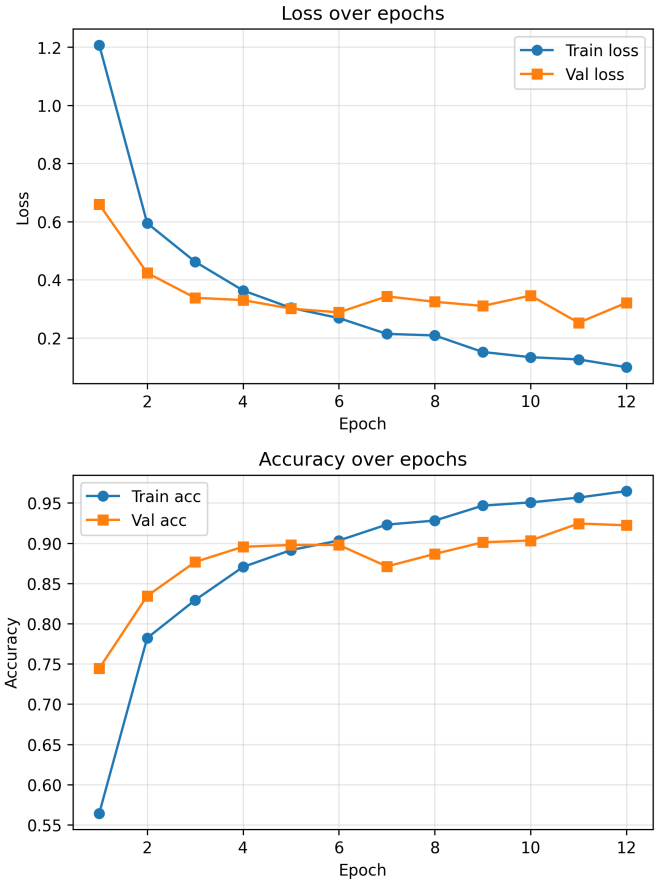


Fig. 1. Loss and accuracy of the from-scratch CNN over training. The validation loss drifts upward in the later epochs, indicating mild overfitting.

TABLE III
PER-CLASS F1 BY MODEL. THE LOWEST SCORES (BOLD) ARE CONSISTENTLY THE TOMATO BLIGHT PAIR.

Class	Scratch	Scr.+Aug	ResNet-18
Tomato healthy	0.985	0.995	1.000
Tomato early blight	0.771	0.816	0.951
Tomato late blight	0.759	0.742	0.930
Tomato bacterial spot	0.905	0.931	0.979
Corn common rust	0.995	1.000	1.000
Corn healthy	0.989	0.989	1.000
Grape black rot	0.892	0.914	0.962
Grape Esca	0.884	0.921	0.967
Potato early blight	0.970	0.941	0.994
Potato late blight	0.897	0.887	0.979
Accuracy	0.910	0.920	0.978

E. The Persistent Tomato Blight Confusion

Table IV looks only at the confusion between tomato early blight and tomato late blight. The total number of such errors did fall as the models improved, going from fourteen for the from-scratch model, to nine with augmentation, to five for ResNet-18. Yet even for ResNet-18, which reached almost ninety-eight per cent overall and cleared up nearly every other confusion, these two classes still had the lowest F1 scores of

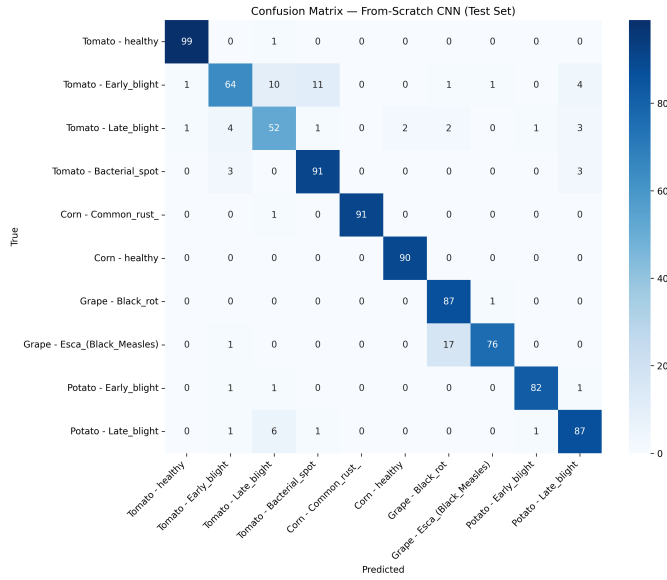


Fig. 2. Confusion matrix for the from-scratch CNN on the test set.

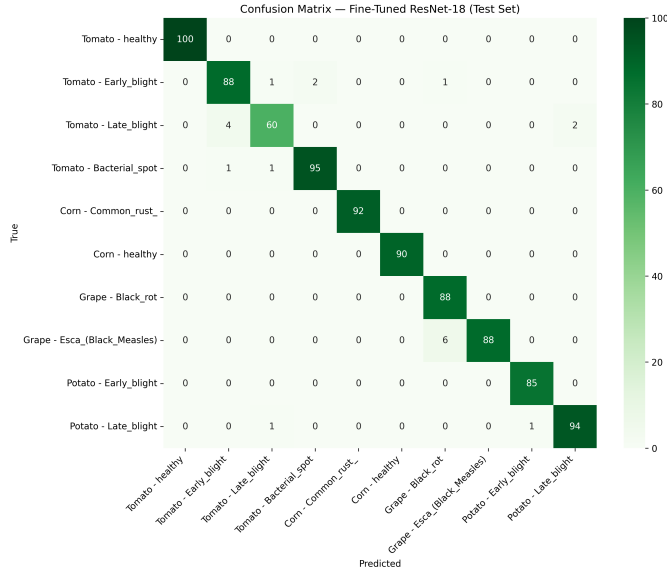


Fig. 3. Confusion matrix for the fine-tuned ResNet-18 on the test set.

all, as Table III shows.

VI. DISCUSSION

Three points stand out from these results. The first is that, on this balanced multi-crop subset, transfer learning helped far more than augmentation did, adding almost seven points of accuracy against roughly one. This fits the idea that the from-scratch model’s main shortcoming was the quality of the features it could learn, rather than a simple lack of variety in the training data. The second point is that augmentation, in our setting, worked mainly as a way to control overfitting: it removed the gap between training and validation performance that we saw without it, but it did not raise accuracy by much.

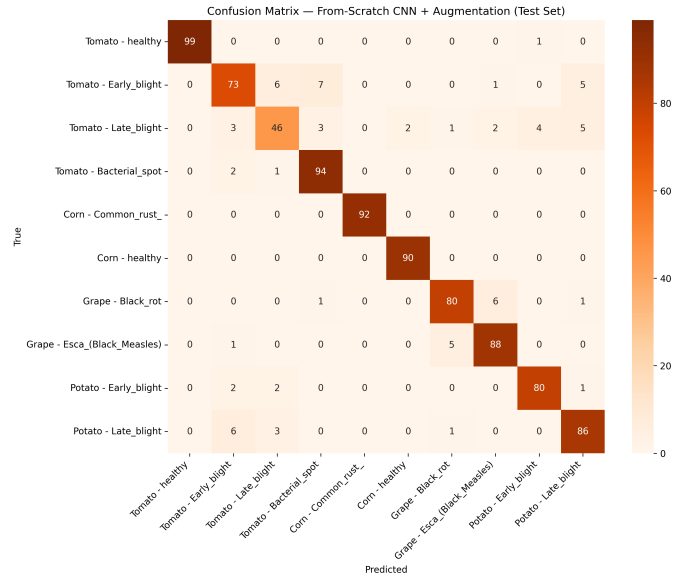


Fig. 4. Confusion matrix for the from-scratch CNN with augmentation on the test set.

TABLE IV
MISCLASSIFICATIONS BETWEEN TOMATO EARLY AND LATE BLIGHT (TEST SET).

Model	E→L	L→E	Total
From-scratch (no aug)	10	4	14
From-scratch (+ aug)	6	3	9
ResNet-18 (transfer)	1	4	5

The third and most important point concerns the two tomato blights. This pair was the hardest for every model. Although transfer learning reduced the number of blight errors, these two classes still had the lowest F1 scores even when the overall accuracy had reached almost ninety-eight per cent, and even though the same model had managed to remove nearly every other confusion. The natural reading of this is that early and late blight on tomato leaves simply look very similar at this image resolution, so that the difficulty lies in the diseases rather than in any weakness of the model. If that is correct, then it has a practical consequence: a tool meant to tell these two diseases apart should be honest about its uncertainty, or should ask for more evidence, rather than committing to a confident single answer that may well be wrong.

VII. LIMITATIONS

This study was deliberately kept small and controlled, and it has several limitations that should be kept in mind. We used only six hundred images per class rather than the full dataset, so our accuracy figures should not be compared directly against results reported on the complete data. The PlantVillage images were all taken against clean backgrounds under controlled conditions, so we cannot say how these models would behave on messy photographs taken in a real field. We also tested only one from-scratch design and one

pretrained network, ResNet-18, and it is possible that a different pretrained model would narrow the remaining blight gap further. Finally, our claim that the blight confusion is intrinsic is an inference drawn from the fact that it persisted, with consistently low F1 scores, across all three models; confirming it directly would require further analysis, for example using a method such as Grad-CAM [8] to see where the model looks, or comparing the model against human experts on the same images.

VIII. CONCLUSION

We compared three ways of training a plant-disease classifier on a balanced ten-class subset of PlantVillage: training a convolutional network from scratch, fine-tuning a pretrained ResNet-18, and retraining the from-scratch network with data augmentation. Transfer learning gave the largest improvement in accuracy and removed most of the confusions between classes, while augmentation mainly served to reduce overfitting. The one confusion that resisted every method was that between tomato early blight and tomato late blight, which stayed the weakest pair in all three models. This points to the two diseases being genuinely similar in appearance rather than to any simple weakness of the models, and it suggests that a practical diagnostic system should treat such pairs with appropriate caution. More broadly, looking beyond a single accuracy figure, to the pattern of errors a model actually makes, gives a more honest account of where these systems can and cannot be trusted.

REPRODUCIBILITY

The code, the exact list of classes, the split seed, and the trained model weights are available at <https://github.com/Syeda-Aliza-Ayaz/Plant-Disease-Classification.git>.

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